RAMANUJAN COLLEGE

UNIVERSITY OF DELHI

COMPUTER NETWORKS

Practicals

SEMESTER - 4

(2025-26)

Submitted by	Submitted to
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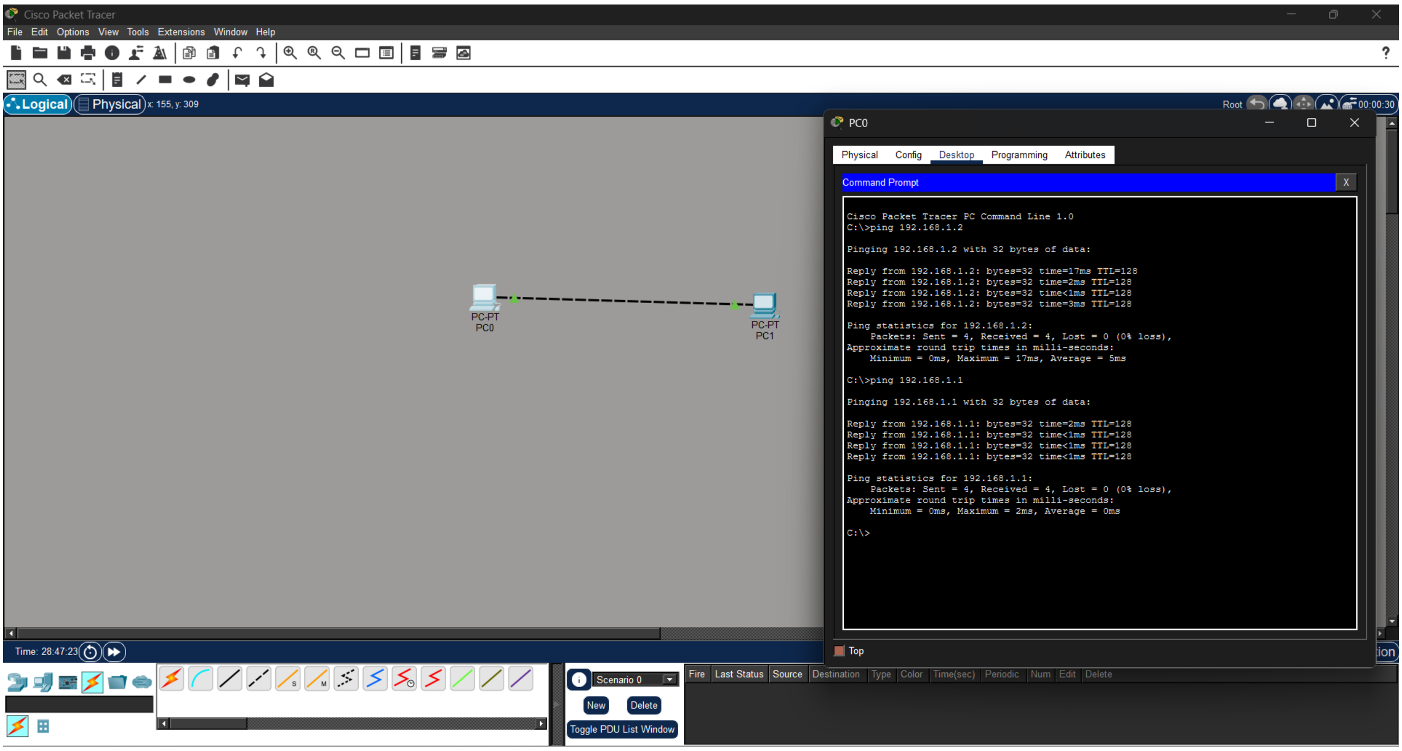
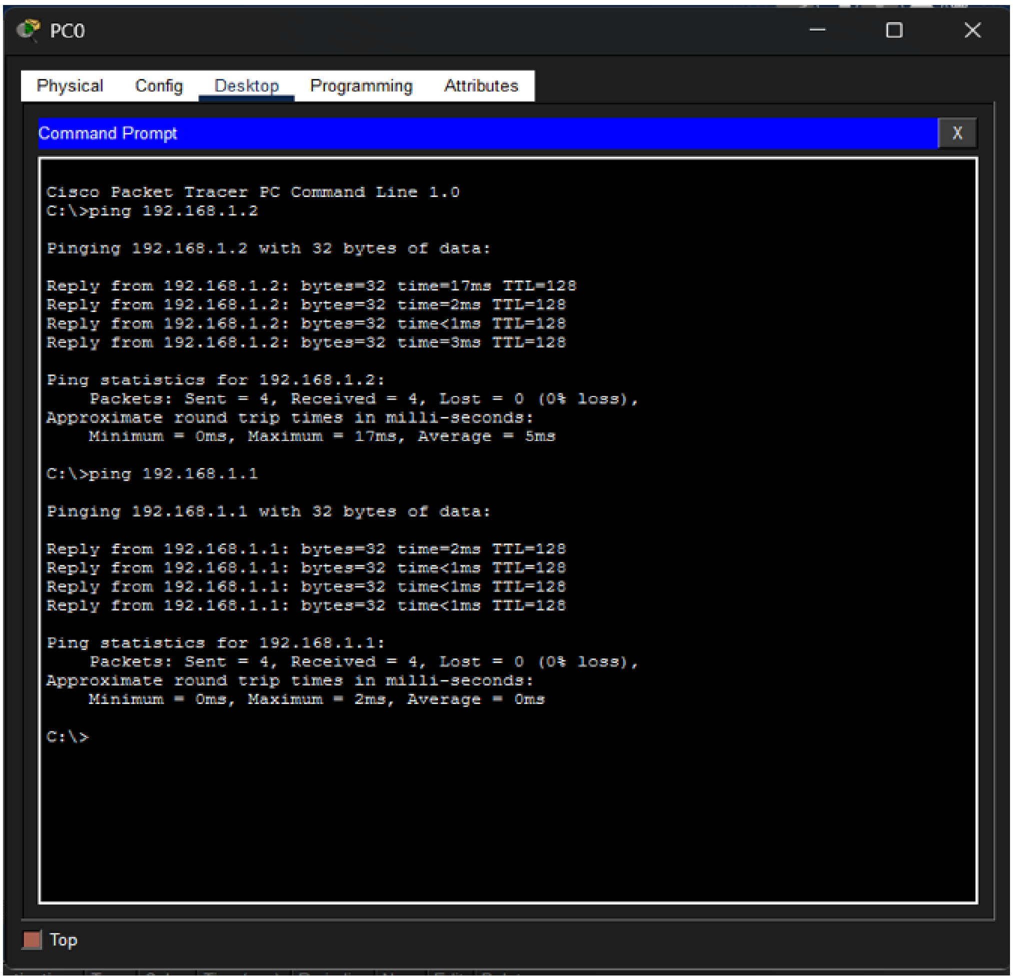
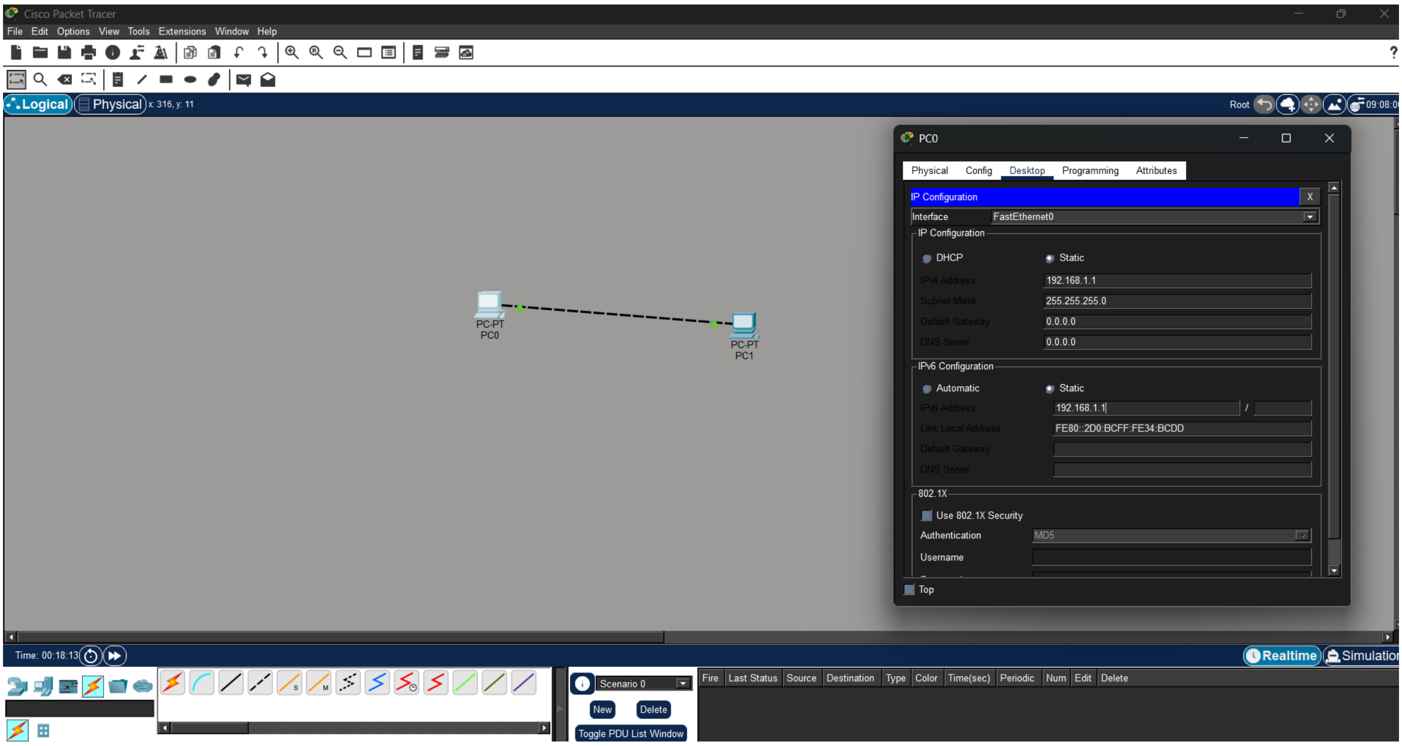
PRACTICALS

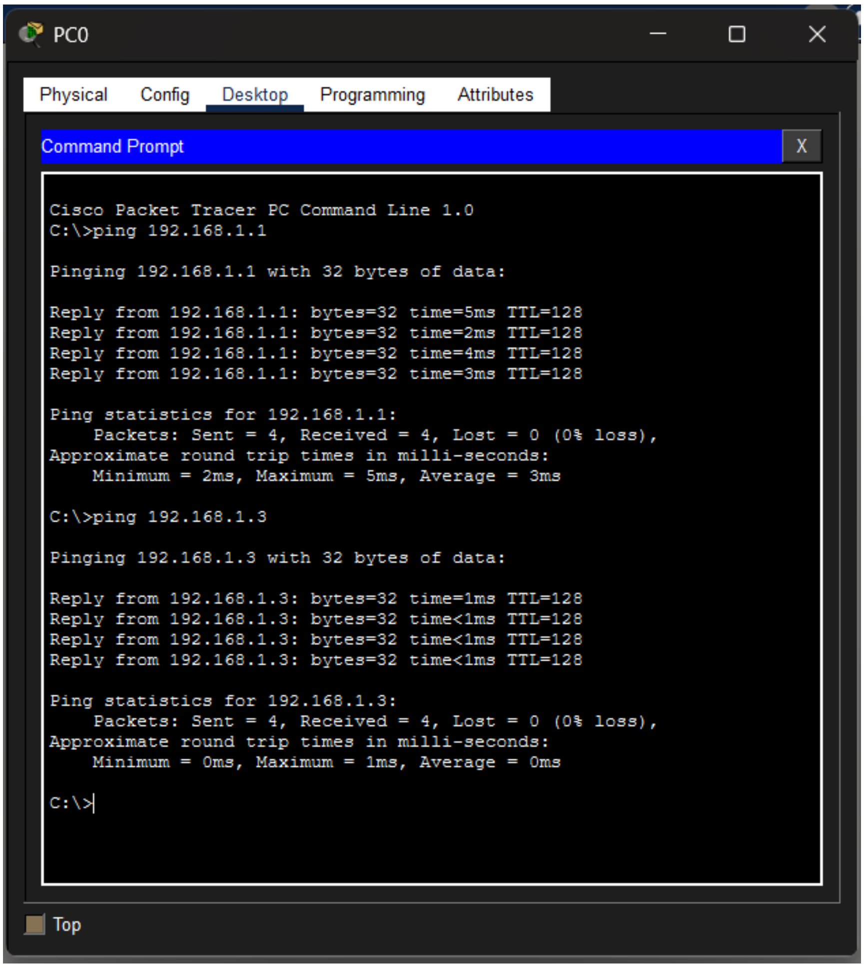
Q1. To Study basic network command and Network configuration commands





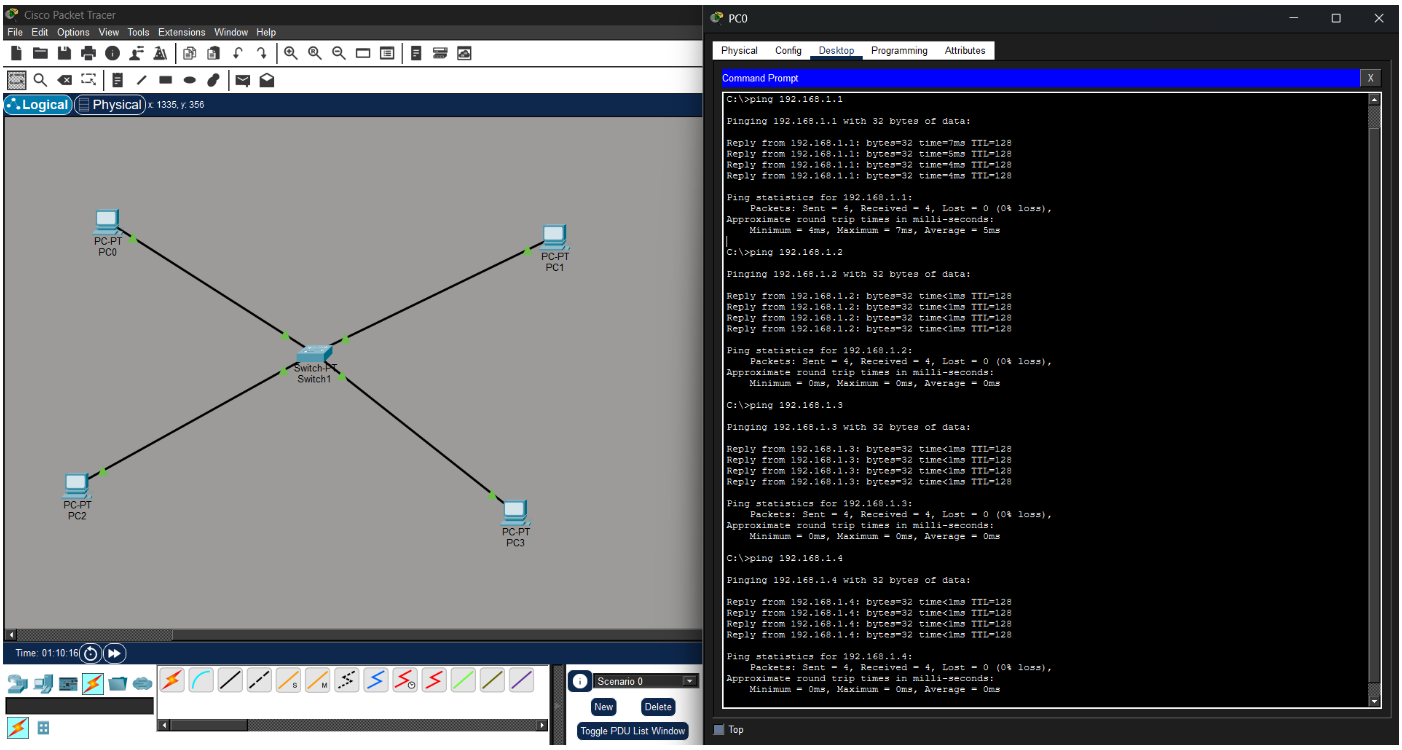
Q2. To study and perform PC to PC communication.

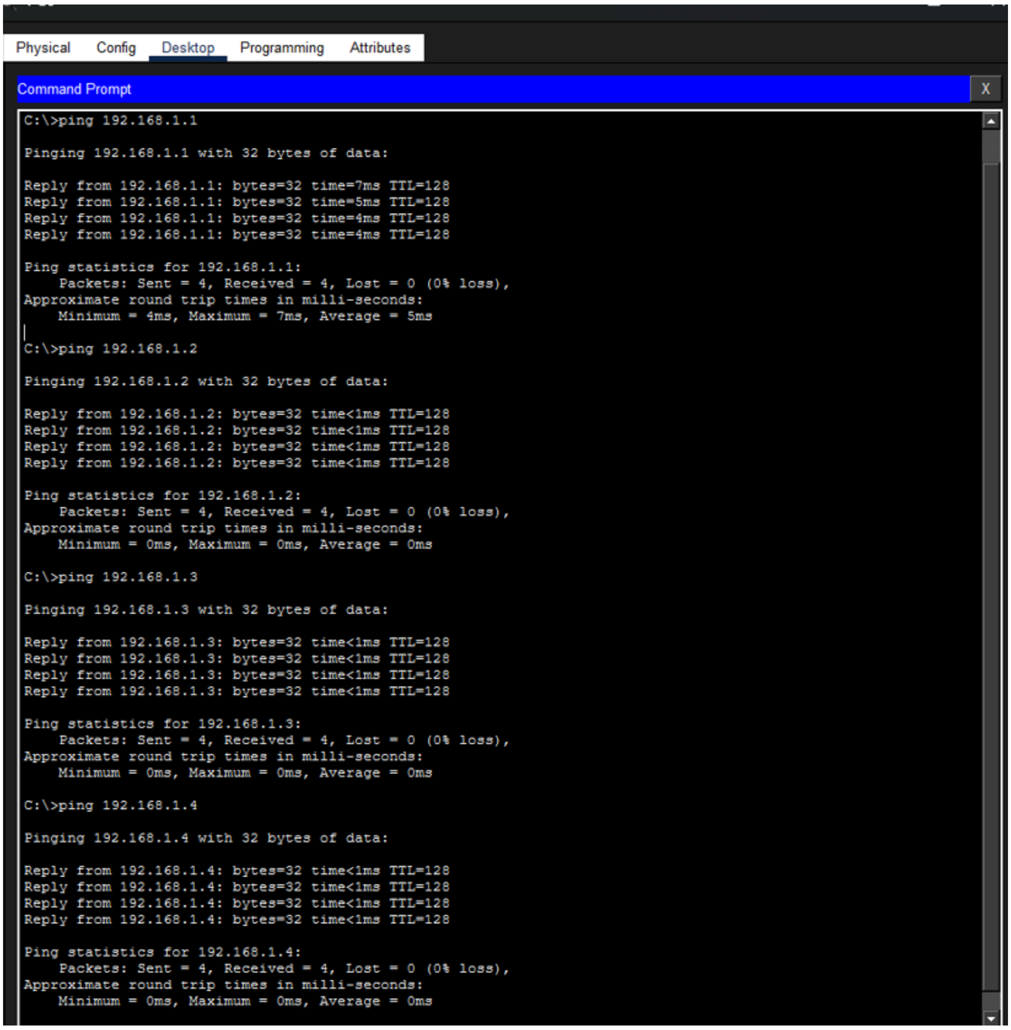




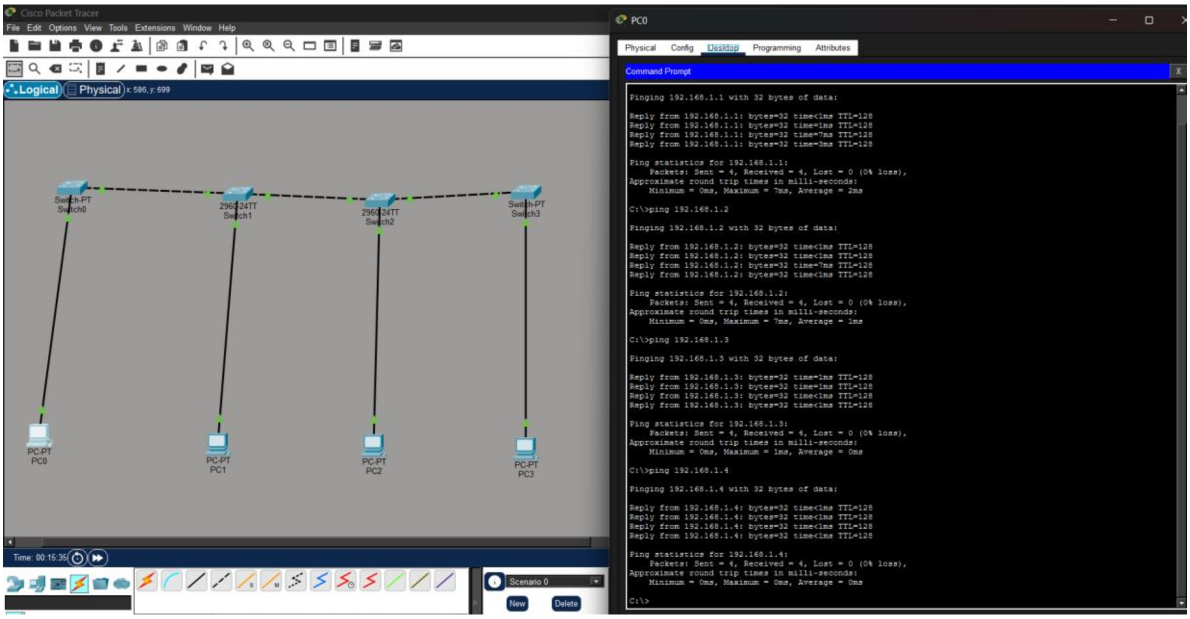

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time=5ms TTL=128
Reply from 192.168.1.1: bytes=32 time=2ms TTL=128
Reply from 192.168.1.1: bytes=32 time=4ms TTL=128
Reply from 192.168.1.1: bytes=32 time=3ms TTL=128

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 5ms, Average = 3ms

C:\>ping 192.168.1.3

Pinging 192.168.1.3 with 32 bytes of data:

Reply from 192.168.1.3: bytes=32 time=1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

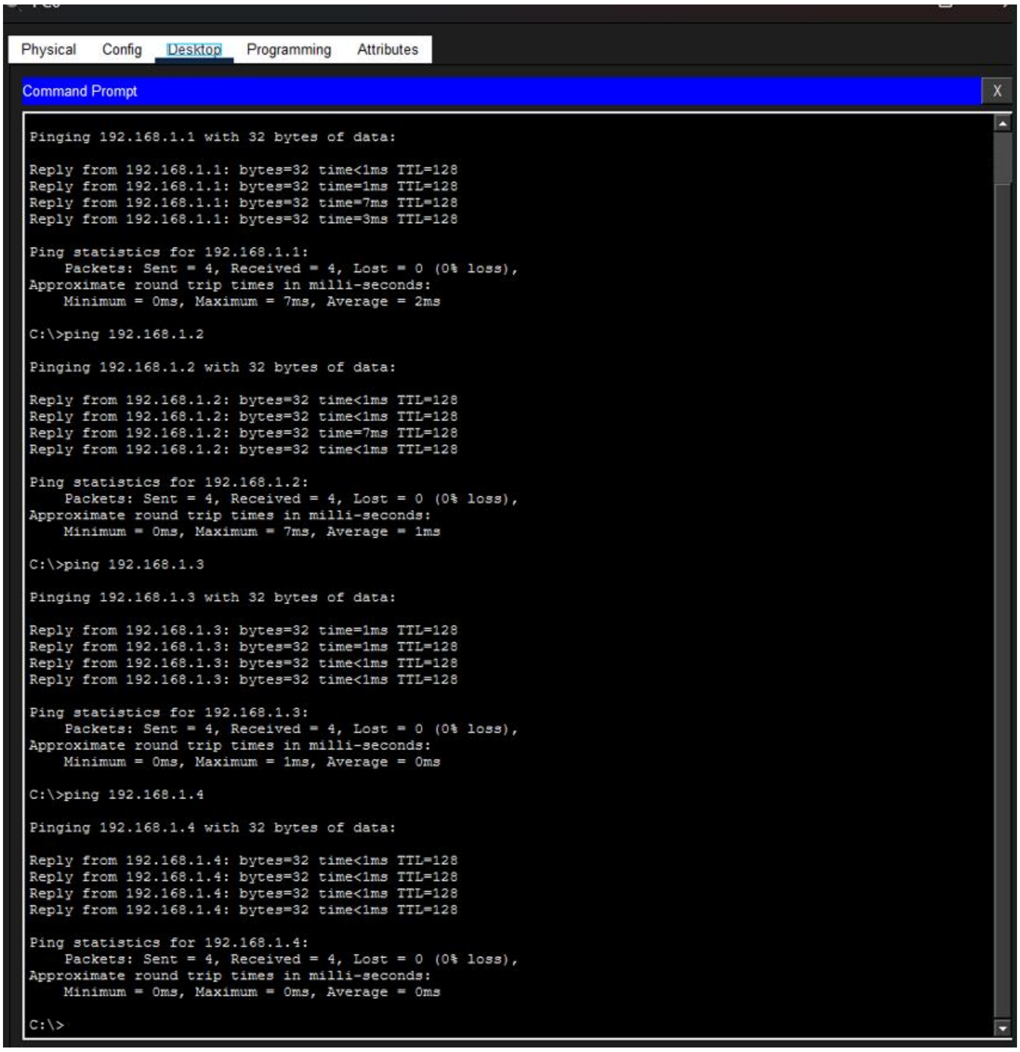
Q3. To create Star topology using Hub and Switch.

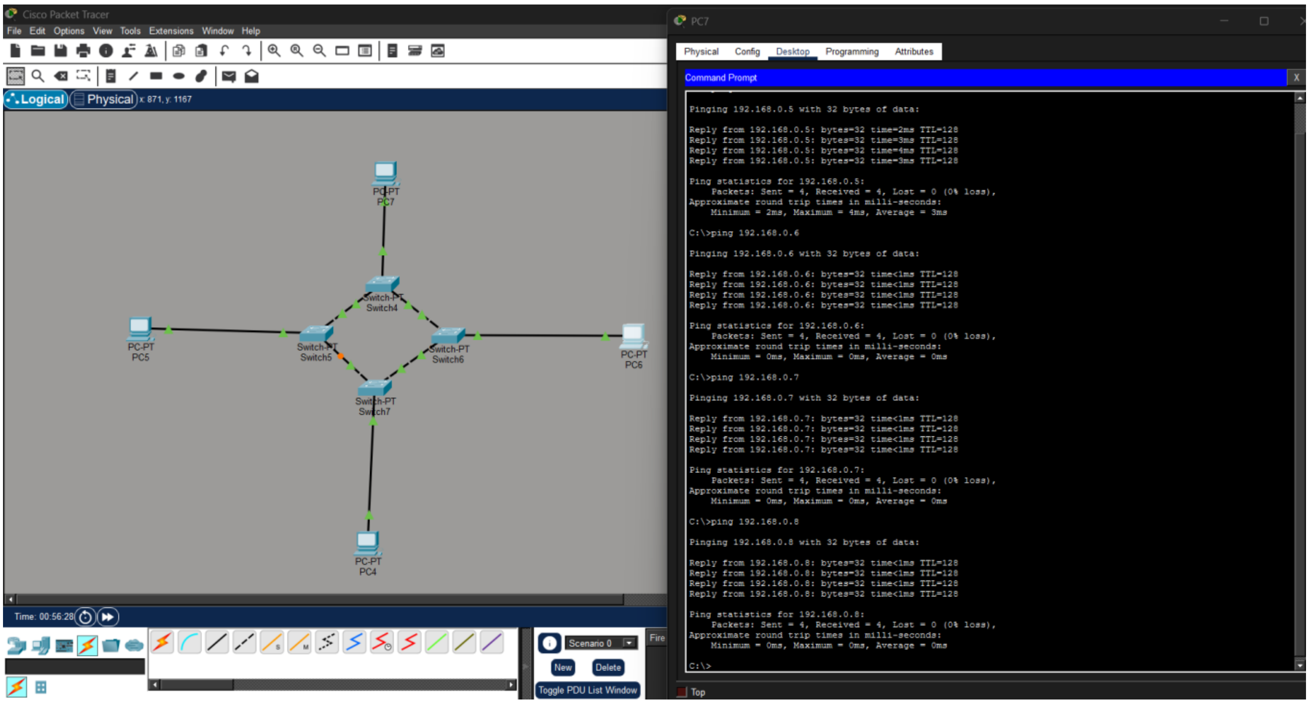


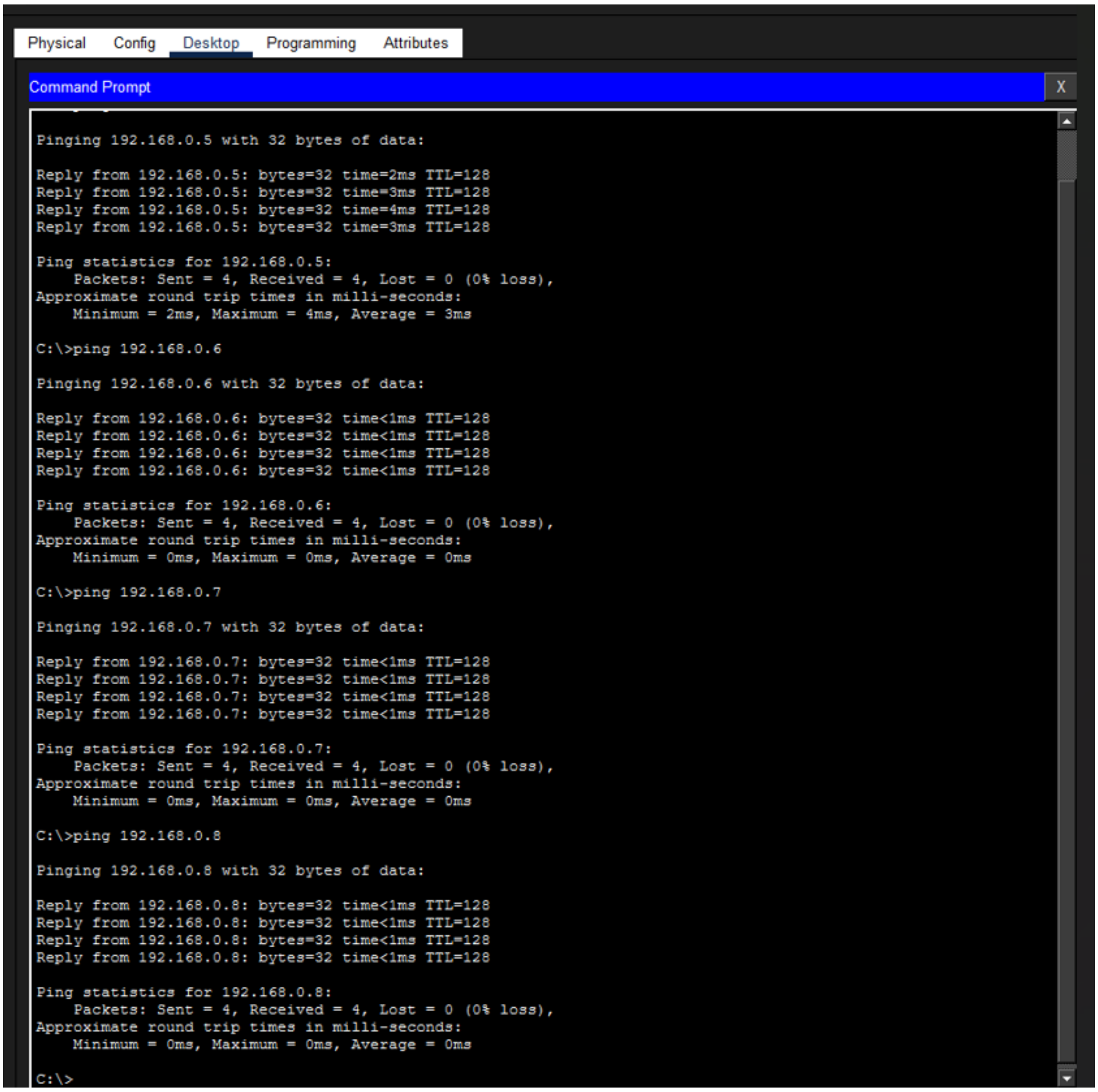


Q4. To create Bus, Ring, Tree, Hybrid, Mesh topologies.





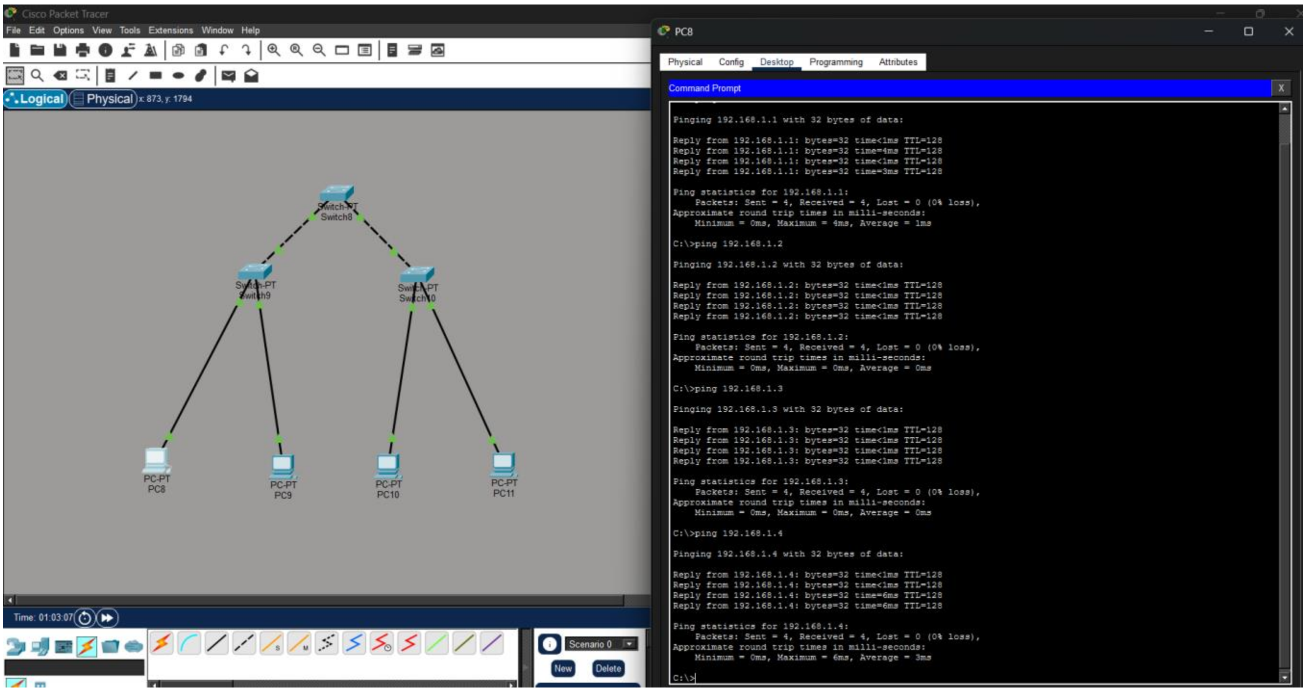


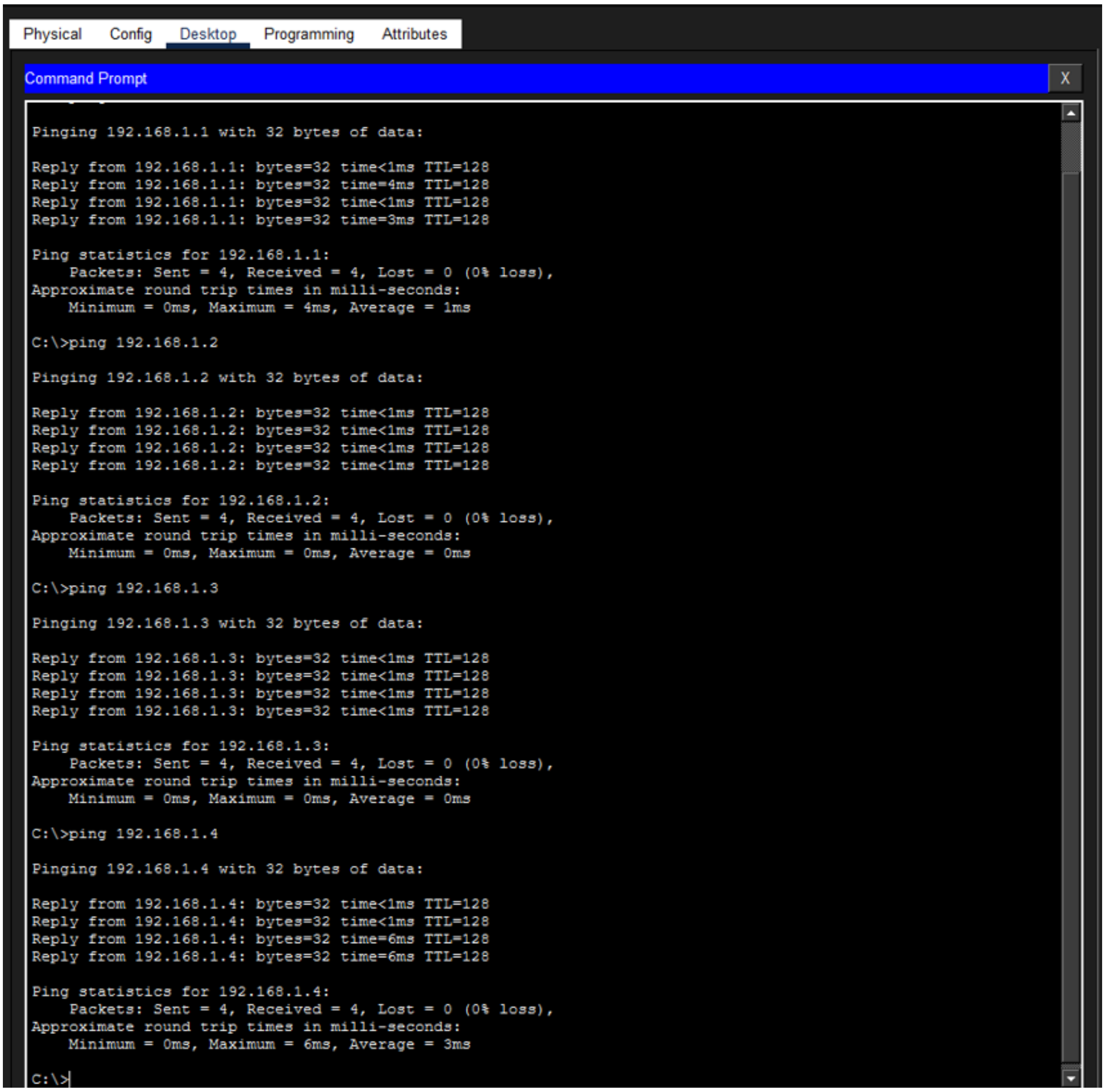


Physical Config Desktop Programming Attributes

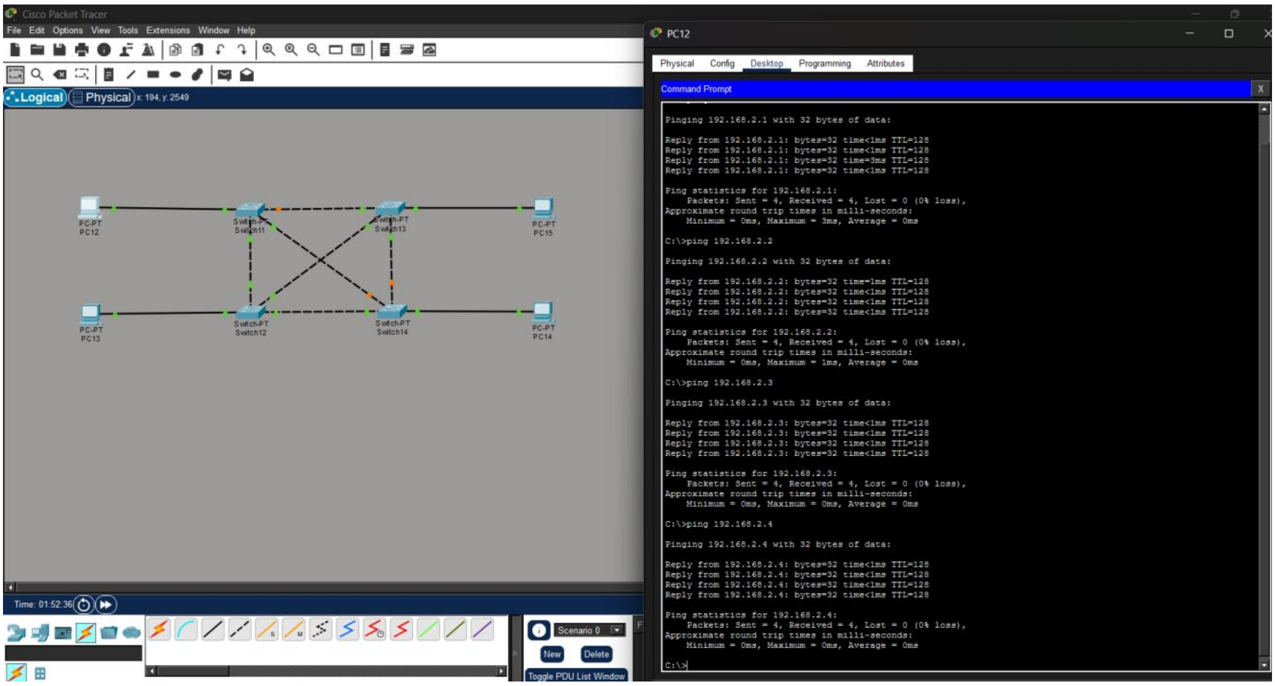
Command Prompt

```
Pinging 192.168.0.5 with 32 bytes of data:

Reply from 192.168.0.5: bytes=32 time=2ms TTL=128
Reply from 192.168.0.5: bytes=32 time=3ms TTL=128
Reply from 192.168.0.5: bytes=32 time=4ms TTL=128
Reply from 192.168.0.5: bytes=32 time=3ms TTL=128

Ping statistics for 192.168.0.5:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 4ms, Average = 3ms

C:\>ping 192.168.0.6

Pinging 192.168.0.6 with 32 bytes of data:

Reply from 192.168.0.6: bytes=32 time<1ms TTL=128
Reply from 192.168.0.6: bytes=32 time<1ms TTL=128
Reply from 192.168.0.6: bytes=32 time<1ms TTL=128
Reply from 192.168.0.6: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.0.6:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.0.7

Pinging 192.168.0.7 with 32 bytes of data:

Reply from 192.168.0.7: bytes=32 time<1ms TTL=128
Reply from 192.168.0.7: bytes=32 time<1ms TTL=128
Reply from 192.168.0.7: bytes=32 time<1ms TTL=128
Reply from 192.168.0.7: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.0.7:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.0.8

Pinging 192.168.0.8 with 32 bytes of data:

Reply from 192.168.0.8: bytes=32 time<1ms TTL=128
Reply from 192.168.0.8: bytes=32 time<1ms TTL=128
Reply from 192.168.0.8: bytes=32 time<1ms TTL=128
Reply from 192.168.0.8: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.0.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

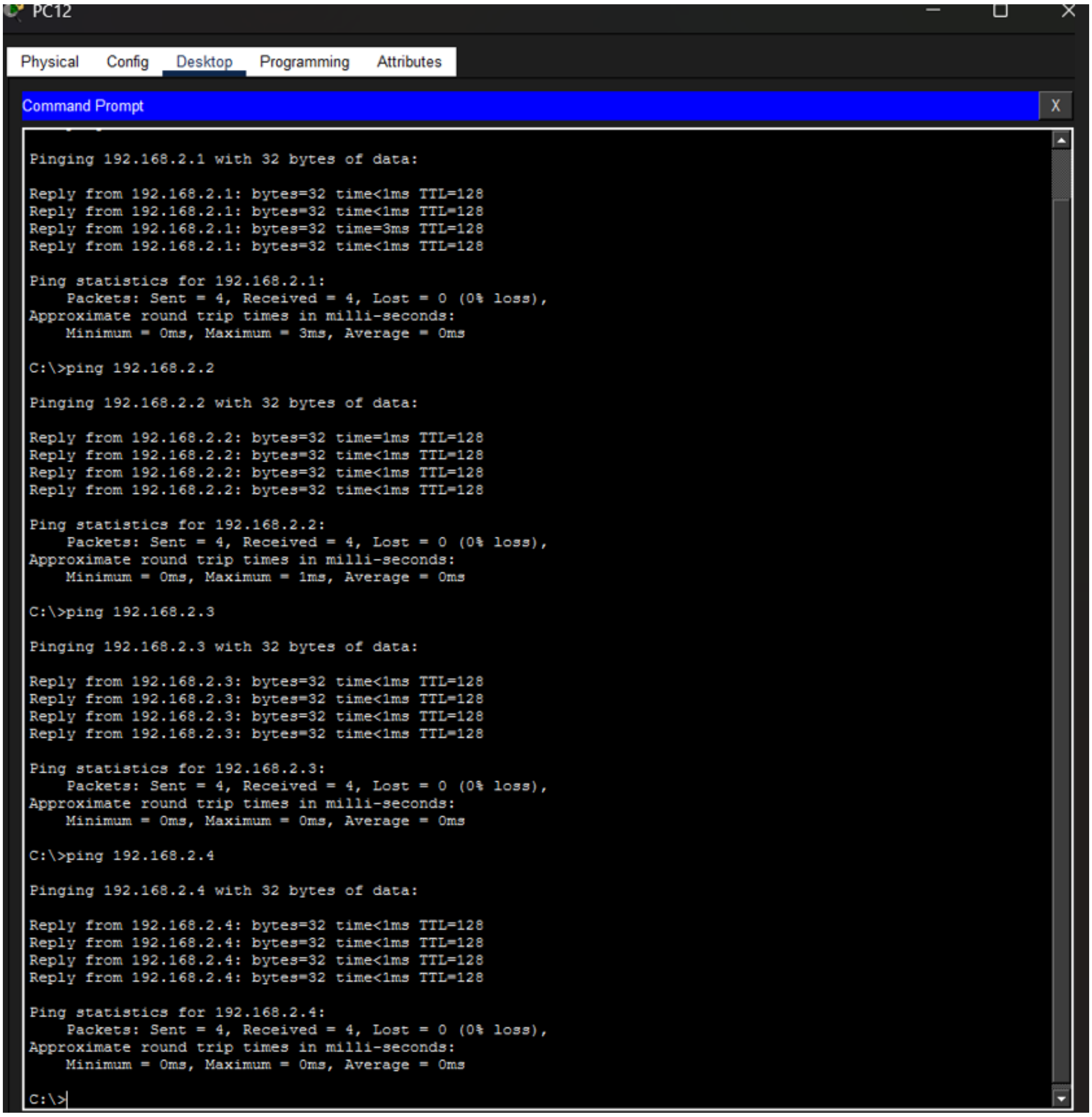




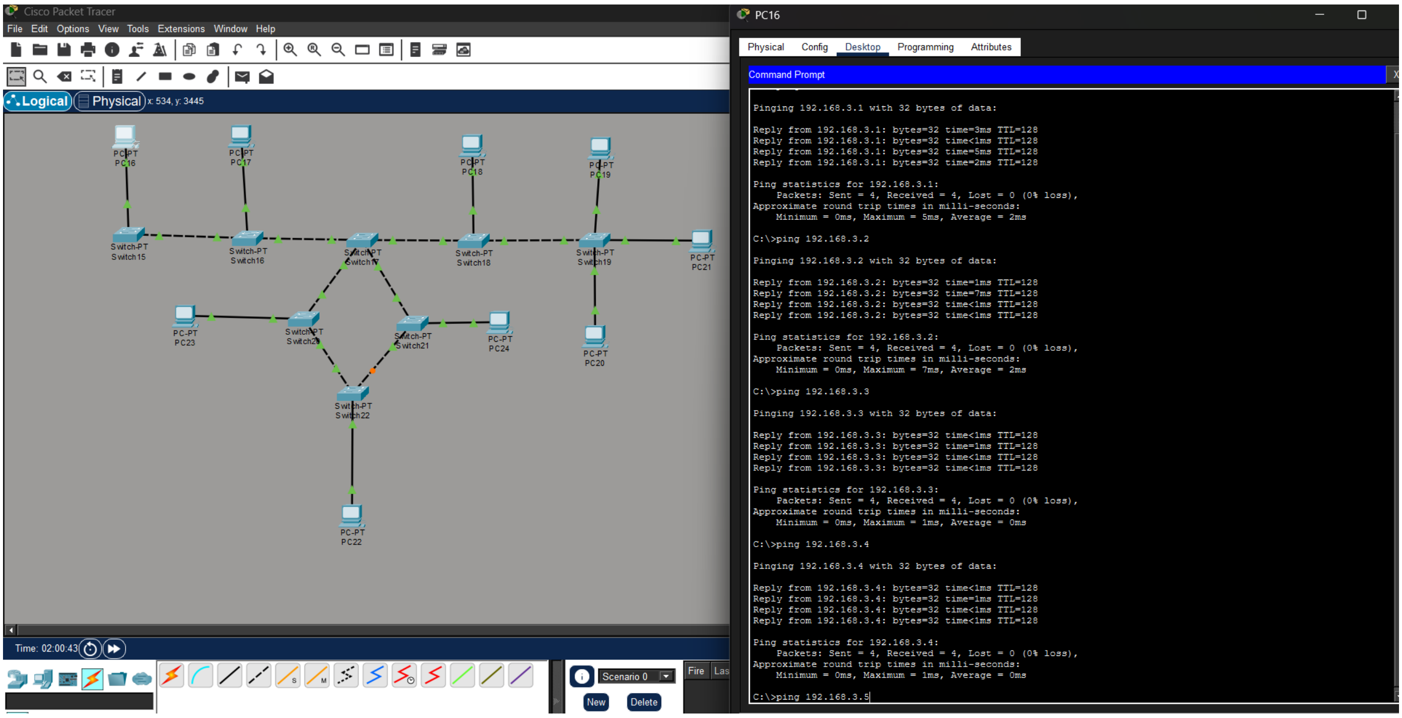

```
Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.1: bytes=32 time=4ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.1: bytes=32 time=3ms TTL=128

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 4ms, Average = 1ms

C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.1.3

Pinging 192.168.1.3 with 32 bytes of data:

Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.1.4

Pinging 192.168.1.4 with 32 bytes of data:

Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time<1ms TTL=128
Reply from 192.168.1.4: bytes=32 time=6ms TTL=128
Reply from 192.168.1.4: bytes=32 time=6ms TTL=128

Ping statistics for 192.168.1.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 6ms, Average = 3ms

C:\>
```

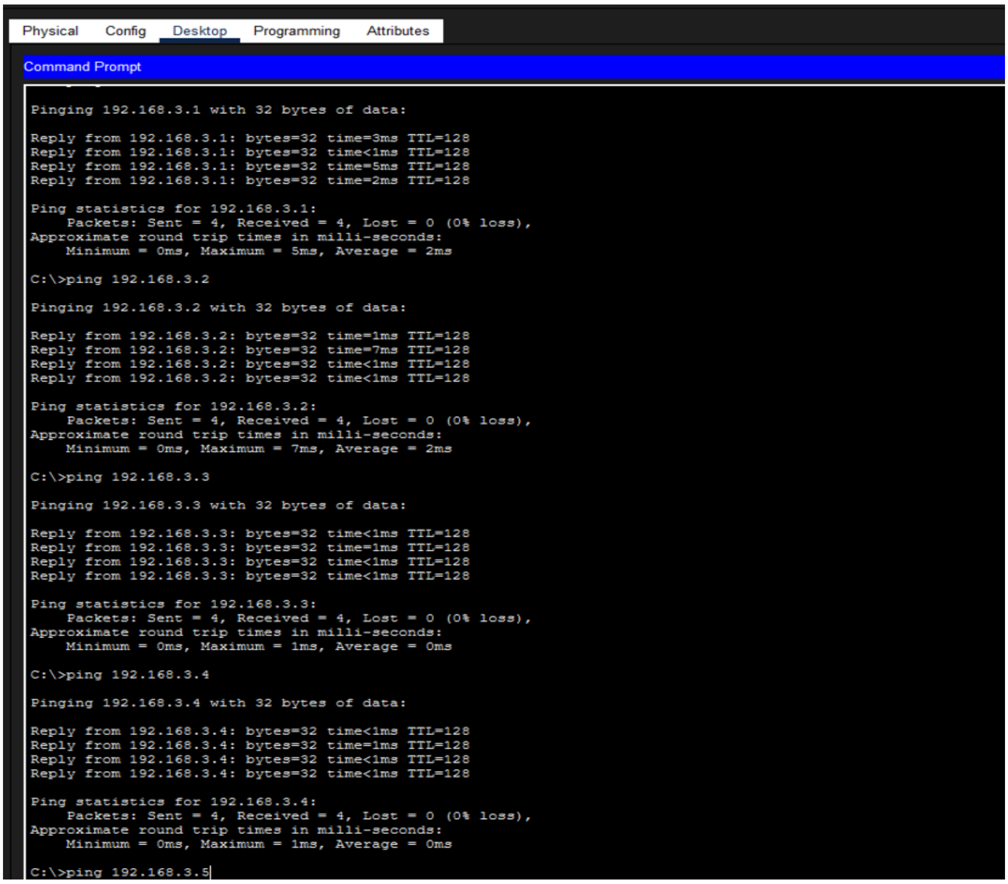


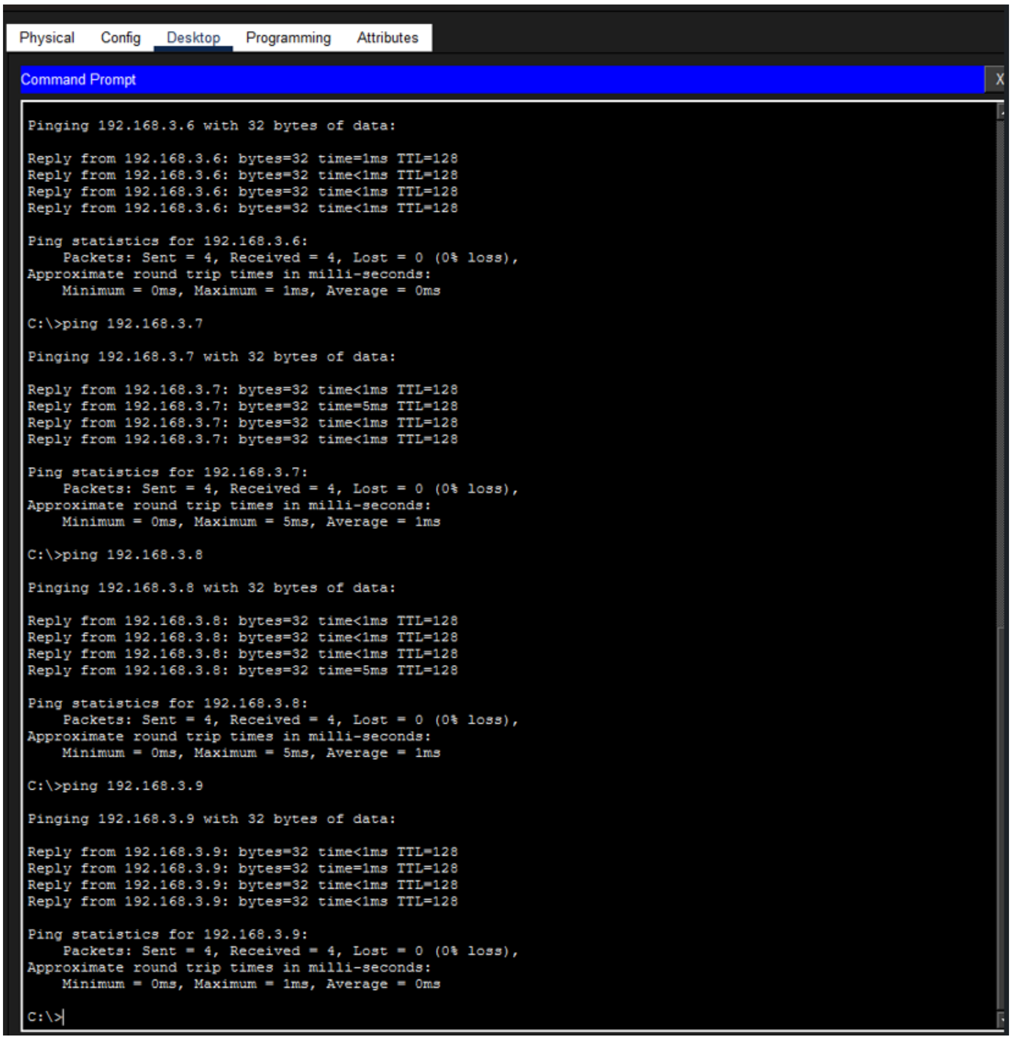
Cisco Packet Tracer
File Edit Options View Tools Extensions Window Help
Logical
Physical
PC-PT
PC12
Switch-PT
Switch11
Switch-PT
Switch13
PC-PT
PC15
PC-PT
PC13
Switch-PT
Switch12
Switch-PT
Switch14
PC-PT
PC14
Scenario 0
New
Delete
Toggle PDU List Window
PC12
Physical
Config
Desktop
Programming
Attributes
Command Prompt

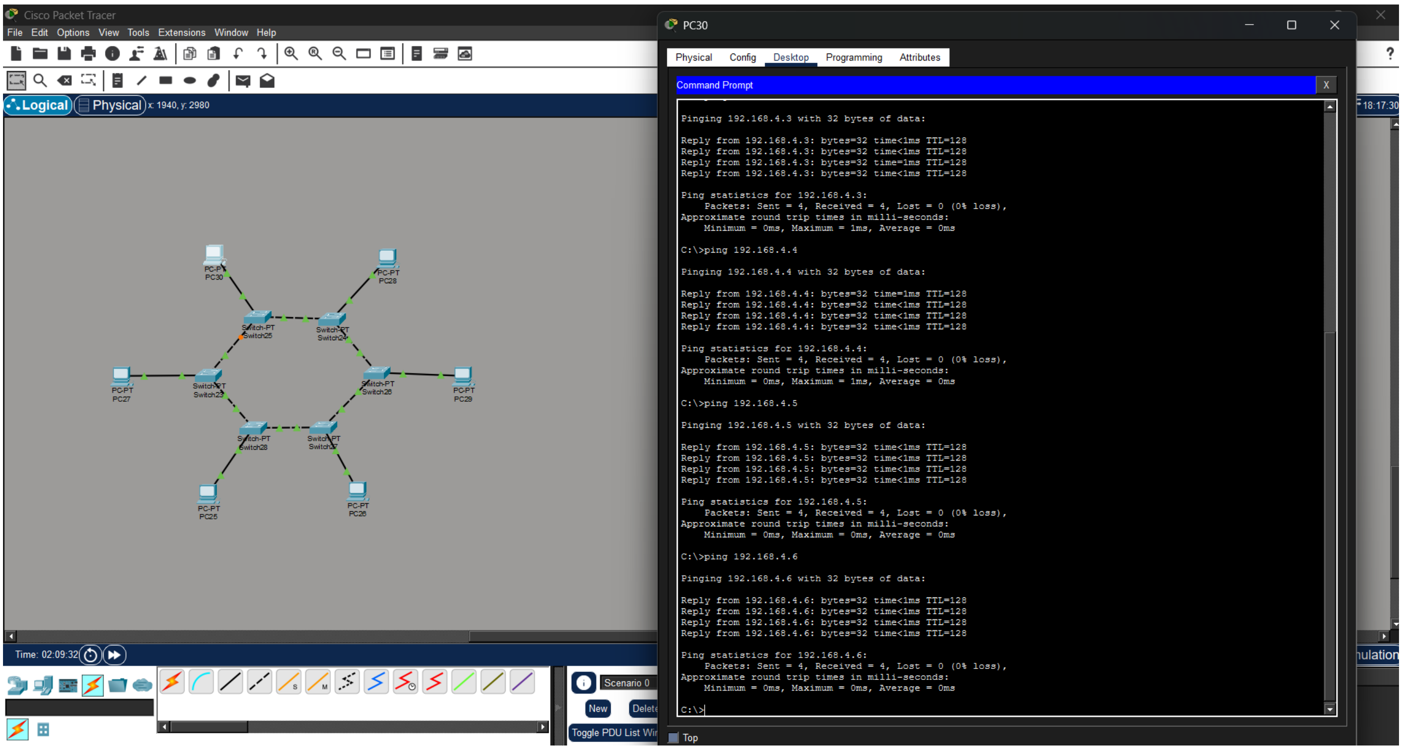


PC12
Physical
Config
Desktop
Programming
Attributes
Command Prompt
X
Pinging 192.168.2.1 with 32 bytes of data:
Reply from 192.168.2.1: bytes=32 time<1ms TTL=128
Reply from 192.168.2.1: bytes=32 time<1ms TTL=128
Reply from 192.168.2.1: bytes=32 time=3ms TTL=128
Reply from 192.168.2.1: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.2.1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 3ms, Average = 0ms
C:\>ping 192.168.2.2
Pinging 192.168.2.2 with 32 bytes of data:
Reply from 192.168.2.2: bytes=32 time=1ms TTL=128
Reply from 192.168.2.2: bytes=32 time<1ms TTL=128
Reply from 192.168.2.2: bytes=32 time<1ms TTL=128
Reply from 192.168.2.2: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.2.2:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>ping 192.168.2.3
Pinging 192.168.2.3 with 32 bytes of data:
Reply from 192.168.2.3: bytes=32 time<1ms TTL=128
Reply from 192.168.2.3: bytes=32 time<1ms TTL=128
Reply from 192.168.2.3: bytes=32 time<1ms TTL=128
Reply from 192.168.2.3: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.2.3:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>ping 192.168.2.4
Pinging 192.168.2.4 with 32 bytes of data:
Reply from 192.168.2.4: bytes=32 time<1ms TTL=128
Reply from 192.168.2.4: bytes=32 time<1ms TTL=128
Reply from 192.168.2.4: bytes=32 time<1ms TTL=128
Reply from 192.168.2.4: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.2.4:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>


Cisco Packet Tracer
File Edit Options View Tools Extensions Window Help
Logical
Physical
x: 534, y: 3445
PC-PT PC16
PC-PT PC17
PC-PT PC18
PC-PT PC19
Switch-PT Switch15
Switch-PT Switch16
Switch-PT Switch17
Switch-PT Switch18
Switch-PT Switch19
PC-PT PC21
PC-PT PC23
Switch-PT Switch20
Switch-PT Switch21
PC-PT PC24
PC-PT PC20
Switch-PT Switch22
PC-PT PC22
Time: 02:00:43
Scenario 0
New
Delete
Fire
PC16
Physical Config Desktop Programming Attributes
Command Prompt


Physical Config Desktop Programming Attributes
Command Prompt
Pinging 192.168.3.1 with 32 bytes of data:
Reply from 192.168.3.1: bytes=32 time=3ms TTL=128
Reply from 192.168.3.1: bytes=32 time<1ms TTL=128
Reply from 192.168.3.1: bytes=32 time=5ms TTL=128
Reply from 192.168.3.1: bytes=32 time=2ms TTL=128
Ping statistics for 192.168.3.1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 5ms, Average = 2ms
C:\>ping 192.168.3.2
Pinging 192.168.3.2 with 32 bytes of data:
Reply from 192.168.3.2: bytes=32 time=1ms TTL=128
Reply from 192.168.3.2: bytes=32 time=7ms TTL=128
Reply from 192.168.3.2: bytes=32 time<1ms TTL=128
Reply from 192.168.3.2: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.3.2:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 7ms, Average = 2ms
C:\>ping 192.168.3.3
Pinging 192.168.3.3 with 32 bytes of data:
Reply from 192.168.3.3: bytes=32 time<1ms TTL=128
Reply from 192.168.3.3: bytes=32 time=1ms TTL=128
Reply from 192.168.3.3: bytes=32 time<1ms TTL=128
Reply from 192.168.3.3: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.3.3:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>ping 192.168.3.4
Pinging 192.168.3.4 with 32 bytes of data:
Reply from 192.168.3.4: bytes=32 time<1ms TTL=128
Reply from 192.168.3.4: bytes=32 time=1ms TTL=128
Reply from 192.168.3.4: bytes=32 time<1ms TTL=128
Reply from 192.168.3.4: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.3.4:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>ping 192.168.3.5


Physical
Config
Desktop
Programming
Attributes
Command Prompt
Pinging 192.168.3.6 with 32 bytes of data:
Reply from 192.168.3.6: bytes=32 time=1ms TTL=128
Reply from 192.168.3.6: bytes=32 time<1ms TTL=128
Reply from 192.168.3.6: bytes=32 time<1ms TTL=128
Reply from 192.168.3.6: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.3.6:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>ping 192.168.3.7
Pinging 192.168.3.7 with 32 bytes of data:
Reply from 192.168.3.7: bytes=32 time<1ms TTL=128
Reply from 192.168.3.7: bytes=32 time=5ms TTL=128
Reply from 192.168.3.7: bytes=32 time<1ms TTL=128
Reply from 192.168.3.7: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.3.7:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 5ms, Average = 1ms
C:\>ping 192.168.3.8
Pinging 192.168.3.8 with 32 bytes of data:
Reply from 192.168.3.8: bytes=32 time<1ms TTL=128
Reply from 192.168.3.8: bytes=32 time<1ms TTL=128
Reply from 192.168.3.8: bytes=32 time<1ms TTL=128
Reply from 192.168.3.8: bytes=32 time=5ms TTL=128
Ping statistics for 192.168.3.8:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 5ms, Average = 1ms
C:\>ping 192.168.3.9
Pinging 192.168.3.9 with 32 bytes of data:
Reply from 192.168.3.9: bytes=32 time<1ms TTL=128
Reply from 192.168.3.9: bytes=32 time=1ms TTL=128
Reply from 192.168.3.9: bytes=32 time<1ms TTL=128
Reply from 192.168.3.9: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.3.9:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>

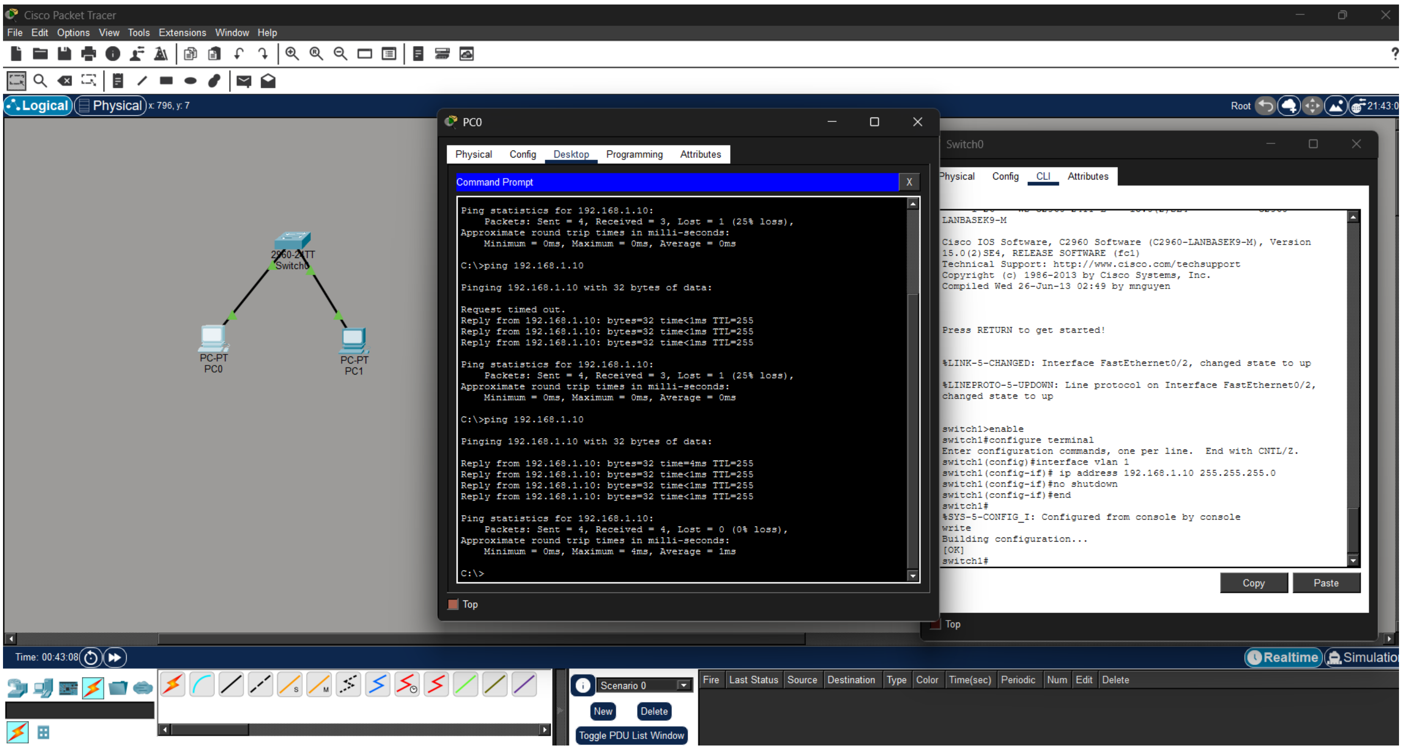

Cisco Packet Tracer
File Edit Options View Tools Extensions Window Help
Logical
Physical
x: 1940, y: 2980
PC-PT PC30
PC-PT PC28
Switch-PT Switch25
Switch-PT Switch24
PC-PT PC27
Switch-PT Switch23
Switch-PT Switch26
PC-PT PC29
Switch-PT Switch28
PC-PT PC25
PC-PT PC26
Time: 02:09:32
Scenario 0
New
Toggle PDU List Wi
PC30
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Pinging 192.168.4.3 with 32 bytes of data:
Reply from 192.168.4.3: bytes=32 time<1ms TTL=128
Reply from 192.168.4.3: bytes=32 time<1ms TTL=128
Reply from 192.168.4.3: bytes=32 time=1ms TTL=128
Reply from 192.168.4.3: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.4.3:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>ping 192.168.4.4
Pinging 192.168.4.4 with 32 bytes of data:
Reply from 192.168.4.4: bytes=32 time=1ms TTL=128
Reply from 192.168.4.4: bytes=32 time<1ms TTL=128
Reply from 192.168.4.4: bytes=32 time<1ms TTL=128
Reply from 192.168.4.4: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.4.4:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>ping 192.168.4.5
Pinging 192.168.4.5 with 32 bytes of data:
Reply from 192.168.4.5: bytes=32 time<1ms TTL=128
Reply from 192.168.4.5: bytes=32 time<1ms TTL=128
Reply from 192.168.4.5: bytes=32 time<1ms TTL=128
Reply from 192.168.4.5: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.4.5:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>ping 192.168.4.6
Pinging 192.168.4.6 with 32 bytes of data:
Reply from 192.168.4.6: bytes=32 time<1ms TTL=128
Reply from 192.168.4.6: bytes=32 time<1ms TTL=128
Reply from 192.168.4.6: bytes=32 time<1ms TTL=128
Reply from 192.168.4.6: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.4.6:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>
Top

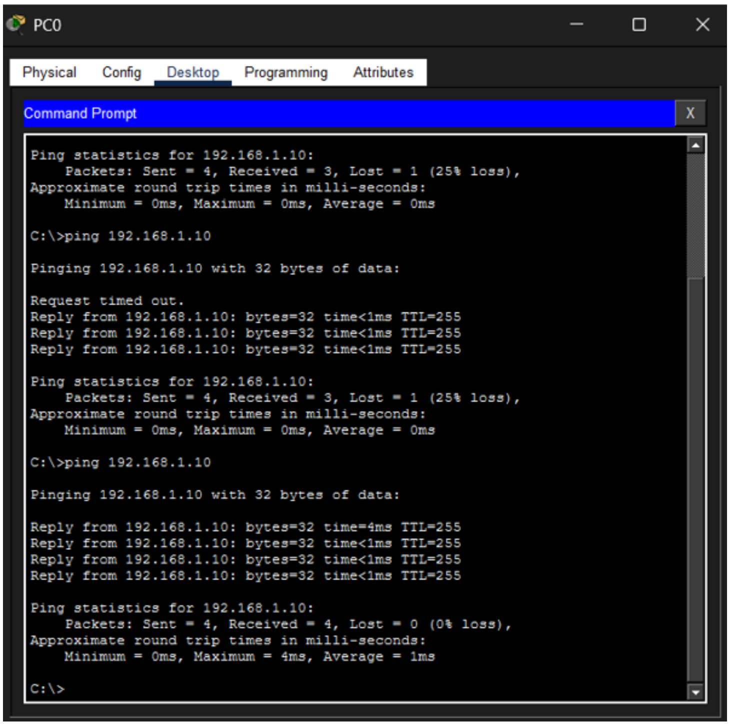
Physical Config Desktop Programming Attributes

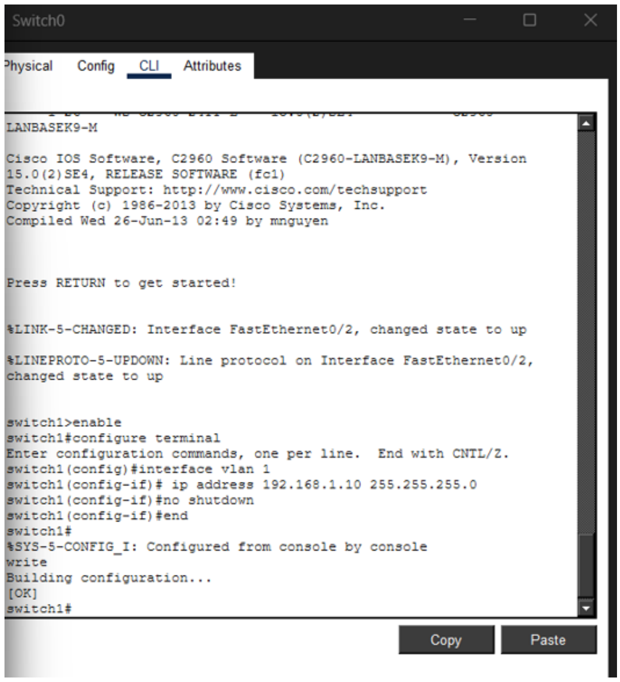
Command Prompt

```
Pinging 192.168.4.3 with 32 bytes of data:

Reply from 192.168.4.3: bytes=32 time<1ms TTL=128
Reply from 192.168.4.3: bytes=32 time<1ms TTL=128
Reply from 192.168.4.3: bytes=32 time=1ms TTL=128
Reply from 192.168.4.3: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.4.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 192.168.4.4

Pinging 192.168.4.4 with 32 bytes of data:

Reply from 192.168.4.4: bytes=32 time=1ms TTL=128
Reply from 192.168.4.4: bytes=32 time<1ms TTL=128
Reply from 192.168.4.4: bytes=32 time<1ms TTL=128
Reply from 192.168.4.4: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.4.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 192.168.4.5

Pinging 192.168.4.5 with 32 bytes of data:

Reply from 192.168.4.5: bytes=32 time<1ms TTL=128
Reply from 192.168.4.5: bytes=32 time<1ms TTL=128
Reply from 192.168.4.5: bytes=32 time<1ms TTL=128
Reply from 192.168.4.5: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.4.5:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 192.168.4.6

Pinging 192.168.4.6 with 32 bytes of data:

Reply from 192.168.4.6: bytes=32 time<1ms TTL=128
Reply from 192.168.4.6: bytes=32 time<1ms TTL=128
Reply from 192.168.4.6: bytes=32 time<1ms TTL=128
Reply from 192.168.4.6: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.4.6:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

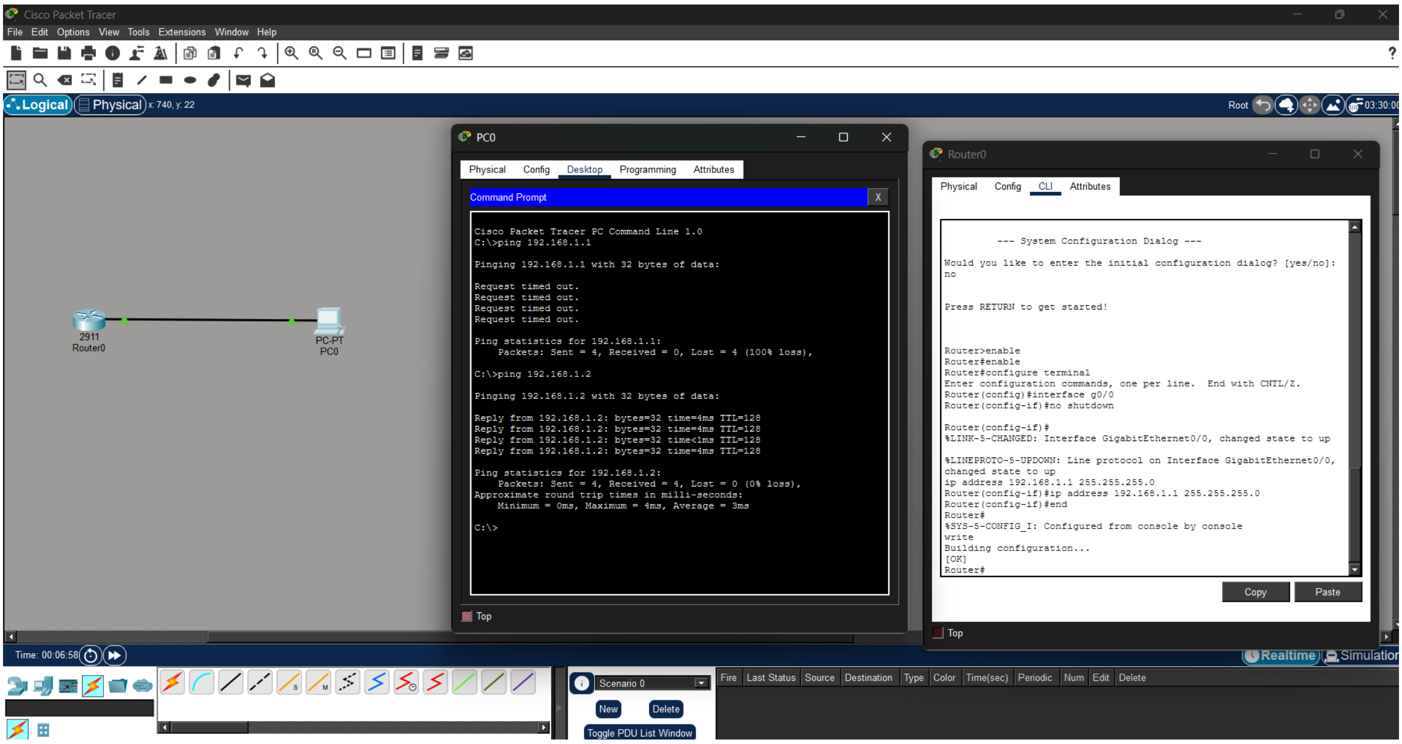
Q5. Perform an initial Switch configuration.

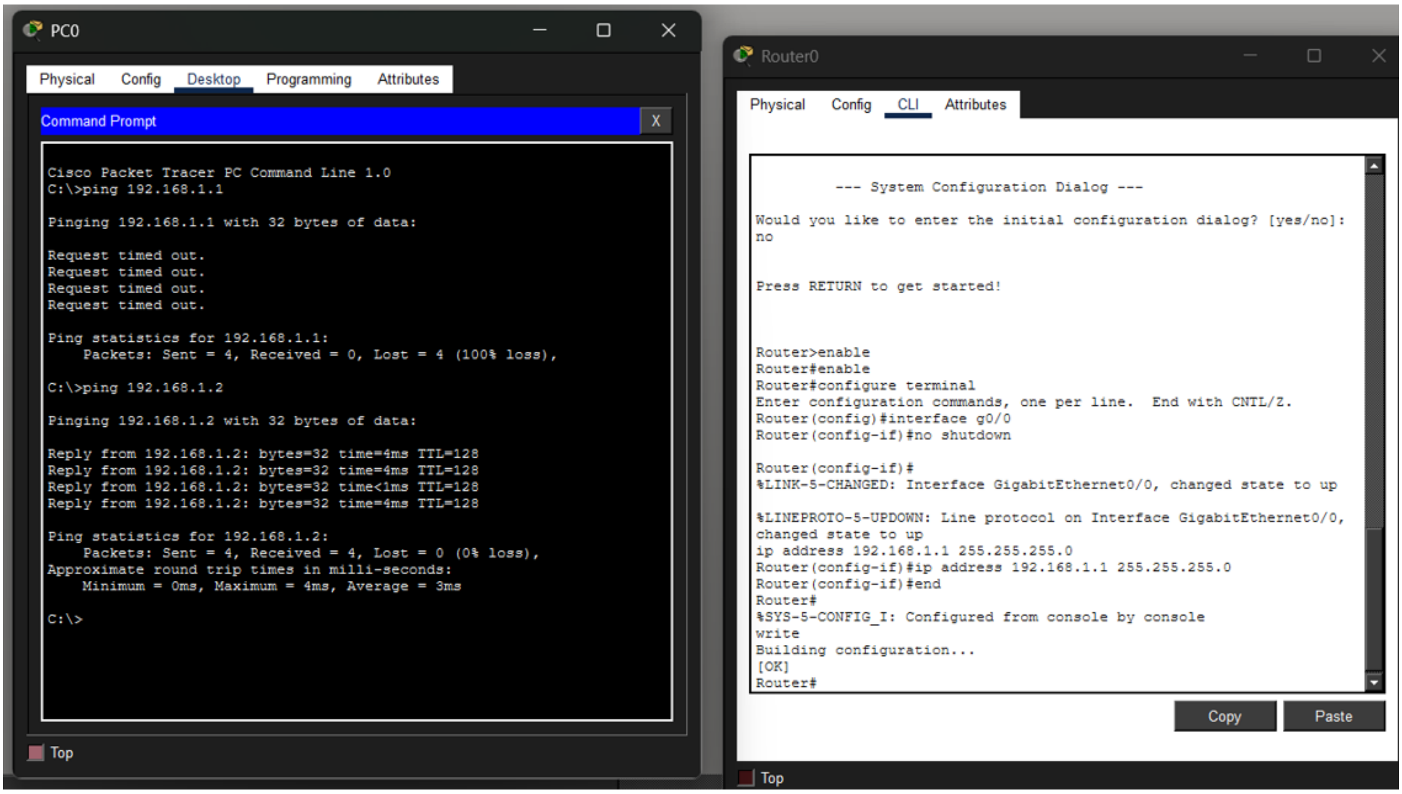




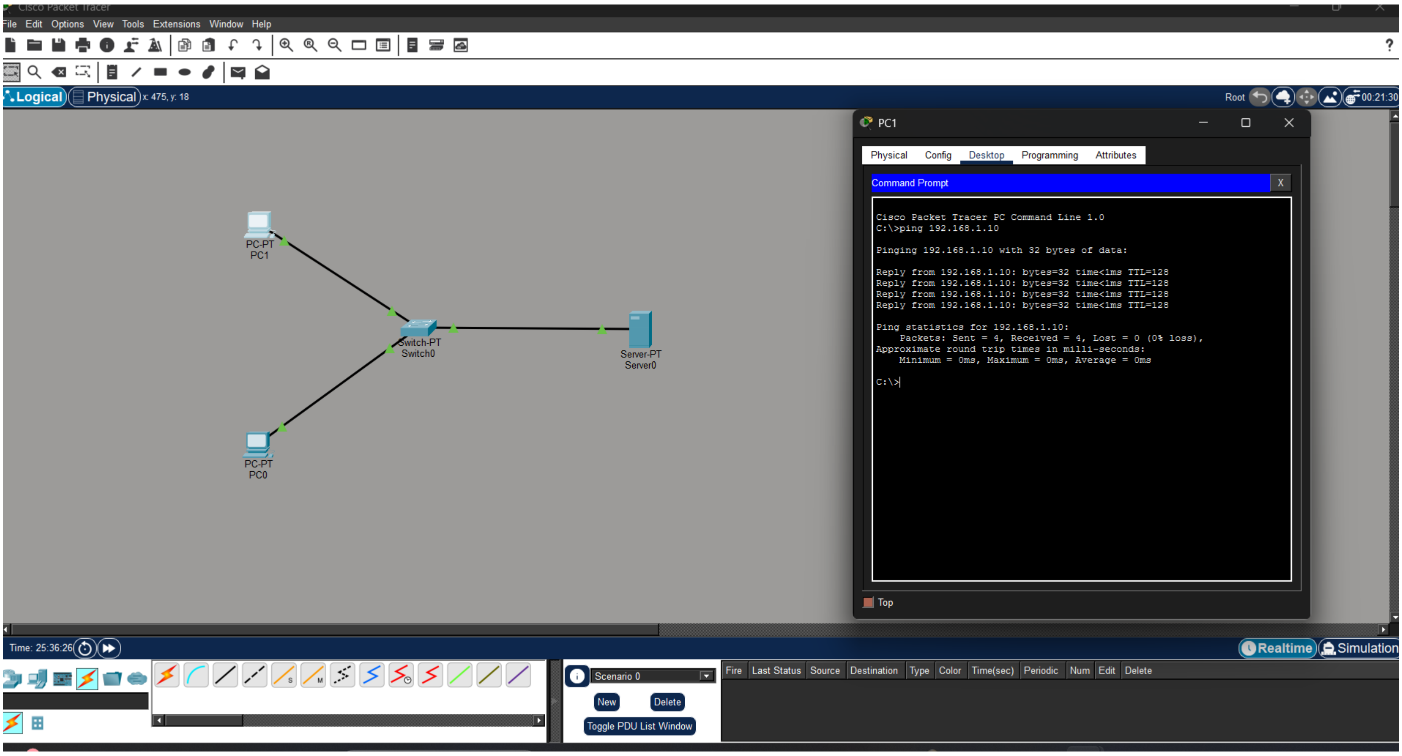


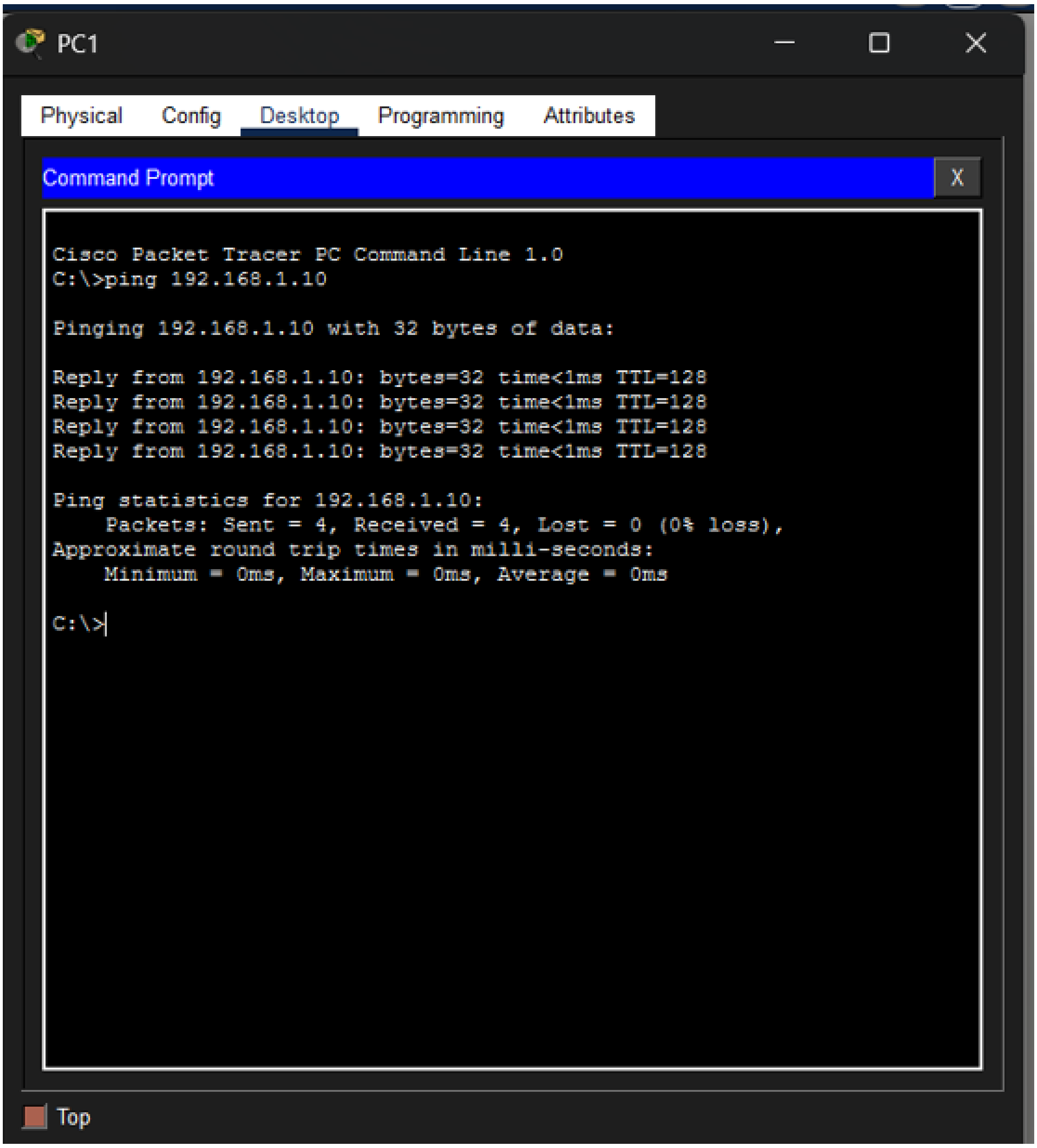
Q6. Perform an initial Router configuration.



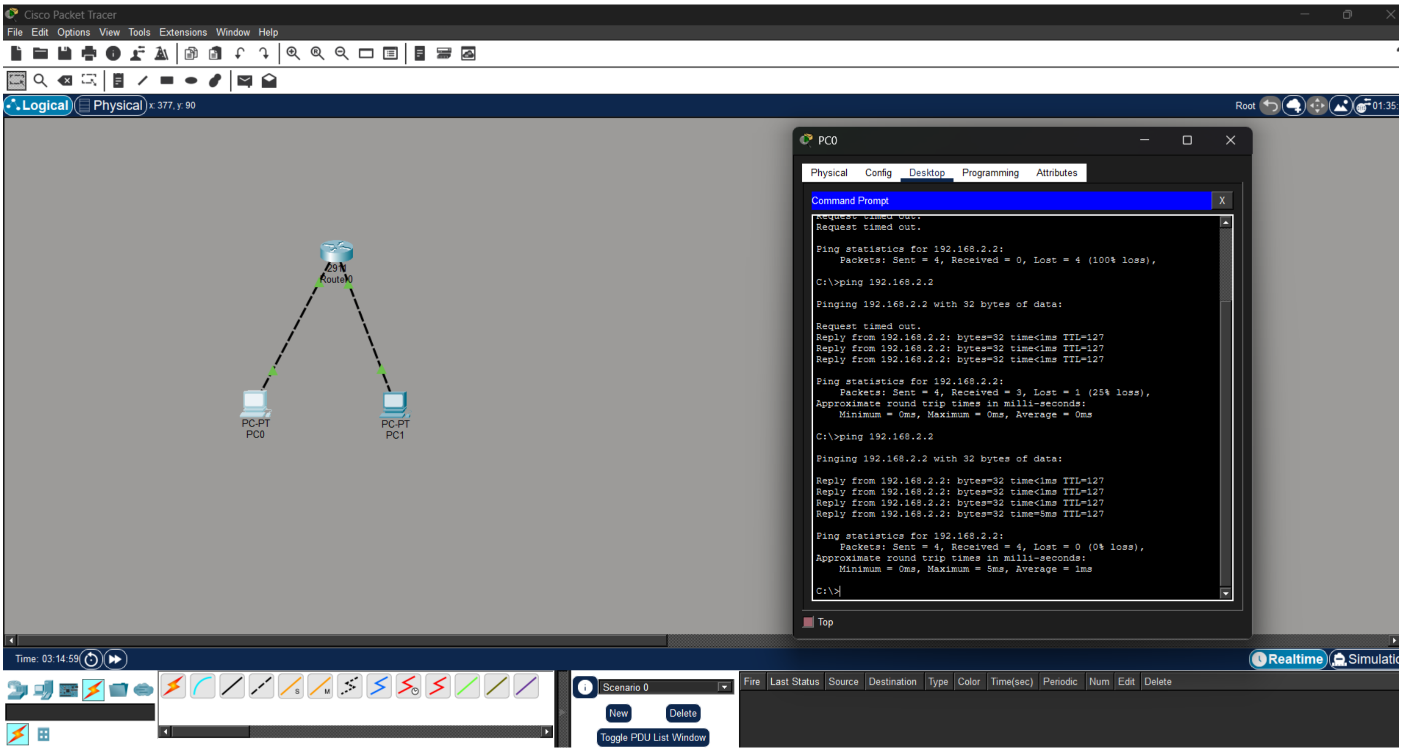


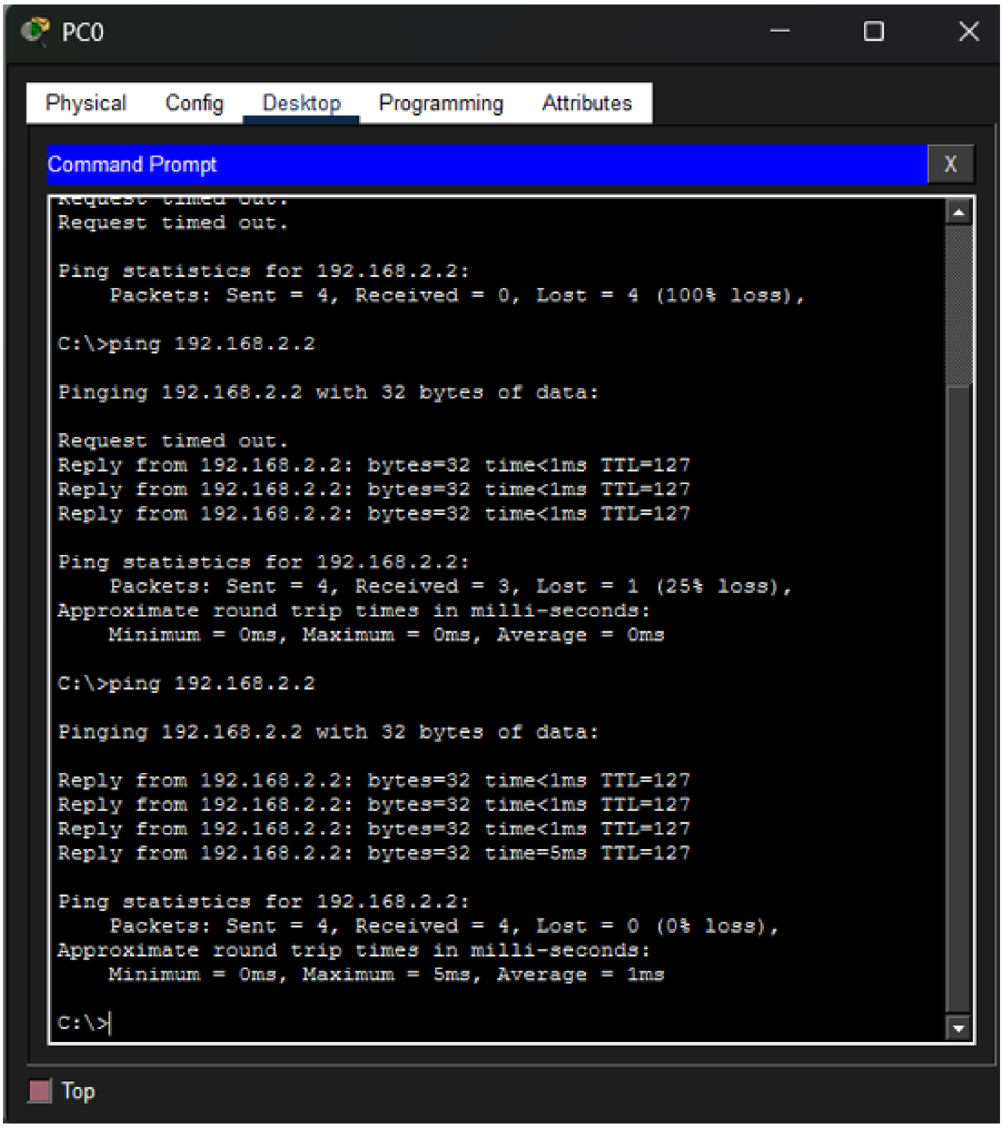
Q7. To implement Client Server Network.





Q8. To implement connection between devices using a router.





Q9. To perform remote desktop sharing within LAN connection.